

ASSESSING ONLINE PRIVACY POLICY COMPLIANCE WITH GDPR AND CCPA
USING NATURAL LANGUAGE PROCESSING.

A Thesis
by
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Abstract

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Privacy policies are the primary mechanism through which organizations communicate data handling practices to users, yet compliance with major regulatory frameworks remains inconsistent and difficult to assess at scale. This study presents an automated natural language processing framework for evaluating privacy policy compliance with the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). Using a BERT-based multi-label classifier trained on 2,571 policies and evaluated against a held-out set of 735, the model achieved an accuracy of 84.5% and an average F1 score of 0.789 across 24 compliance elements. Analysis of 3,674 policies drawn from the Tranco top 100,000 domains revealed a mean GDPR compliance of 68.9% compared to 42.6% for CCPA, a statistically significant gap driven largely by organizations extending GDPR disclosure templates without addressing CCPA-specific obligations. Critically under addressed requirements include notice at collection (10.3%), non-discrimination rights (16.1%), and the right to delete (20.5%). Compliance deficits were found to be systemic and sector-agnostic, with no statistically significant variation across finance, health, retail, technology, and media. The findings demonstrate that automated NLP assessment is a viable and scalable approach to

large-scale compliance auditing, and that dual-framework compliance cannot be achieved through a single unified disclosure strategy.

Keywords: Privacy policy compliance, GDPR, CCPA, Natural language processing, BERT, Multi-label classification, Automated compliance assessment

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Dedication

This thesis is dedicated to my family and to everyone who supported and encouraged me throughout this journey. Your belief in me made this accomplishment possible.

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Foreword

This thesis is submitted in partial fulfillment of the requirements for the Master of Science degree at Appalachian State University. The work presented in this document reflects my original research on assessing online privacy policy compliance with the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) using natural language processing methods.

AI Disclosure Statement

I used Napkin AI (<https://www.napkin.ai/>) to design the research design diagram and the model architecture diagram included in this thesis. The tool was used only for visual diagram generation. All research design decisions, methodology, analysis, interpretations, and written content were developed by me, and the generated visuals were reviewed and edited to accurately represent my work.

Chapter 1

Introduction

Privacy policies are the primary mechanism through which organizations communicate how they collect, use, share, and protect user data. With the introduction of comprehensive privacy regulations such as the European Union's General Data Protection Regulation (GDPR) (European Union, 2016/2018) and the California Consumer Privacy Act (CCPA) (California Legislature, 2018/2020), organizations are under increasing pressure to ensure their privacy policies satisfy strict legal requirements. However, manually verifying compliance across thousands of websites is impractical, making automated approaches necessary for large-scale compliance assessment (Al Bassit et al., 2021).

Background and Context

Privacy policies have become significantly more complex over time. Wagner's (2022) longitudinal study of privacy policies between 1996 and 2021 found that policies have increased substantially in both length and complexity. While early privacy policies were often short and relatively easy to understand, modern policies are typically lengthy and written in highly legalistic language.

This growing complexity is driven by the rapid expansion of digital services, the increased sophistication of data collection technologies, and the need for organizations to comply with multiple privacy regulations across different jurisdictions. As businesses increasingly operate internationally, privacy compliance has become a multifaceted challenge involving overlapping and sometimes conflicting legal obligations.

Two of the most influential privacy regulatory frameworks shaping compliance requirements are the GDPR and the CCPA. The GDPR is widely regarded as one of the most

comprehensive privacy laws, imposing explicit transparency obligations regarding personal data processing, particularly under Articles 13 and 14 (European Union, 2016/2018). It also grants individuals enforceable rights, including the rights to access, rectify, erase, and transfer personal data under Articles 15 through 20 (European Union, 2016/2018). Furthermore, GDPR emphasizes accountability through requirements such as Data Protection Impact Assessments (DPIAs) under Article 35, and strict conditions for lawful processing and consent under Articles 6 and 7 (European Union, 2016/2018).

Similarly, the CCPA reflects several GDPR-aligned principles, but introduces requirements specific to California's legal framework. These include the consumer's right to know what personal information is collected and sold (Section 1798.100), the right to opt out of the sale of personal information (Section 1798.120), and the right to request deletion of personal information under certain exceptions (Section 1798.105) (California Legislature, 2018/2020). The CCPA also includes non-discrimination protections (Section 1798.125), ensuring that consumers are not penalized for exercising privacy rights (California Legislature, 2018/2020). Together, these regulations illustrate the increasing complexity of privacy governance and the rising expectations placed on organizations to provide transparent and legally compliant privacy disclosures.

Manual review of privacy policies presents several limitations. Privacy policies often contain vague or ambiguous language, requiring expert interpretation to identify explicit and implicit disclosures (Olukoya, 2022; Reidenberg et al., 2016). This process is highly time-consuming and cannot scale effectively. Wilson et al. (2016) note that manual annotation is extremely labor-intensive, making it unrealistic to apply across the thousands of websites potentially subject to GDPR or CCPA obligations.

Additionally, even trained annotators frequently disagree in their interpretations, leading to inconsistencies and inter-rater variability (Wang et al., 2025). Privacy policies are also dynamic documents, frequently updated in response to organizational changes, new technologies, or evolving regulatory interpretations. Ehimuan et al. (2024) emphasize that continuous monitoring is necessary, as one-time compliance assessments quickly become outdated.

Problem Statement

Despite the growing importance of privacy policy compliance, there remains a significant gap in automated tools capable of systematically evaluating whether privacy policies satisfy the legal disclosure requirements of major data protection regulations, particularly GDPR (European Union, 2016/2018) and CCPA (California Legislature, 2018/2020). This gap is increasingly problematic because many modern digital services operate across borders, requiring organizations to comply with multiple regulatory frameworks simultaneously. Lim and Oh (2025) highlight that organizations must “simultaneously comply with overlapping data protection regimes when their services extend across multiple jurisdictions.” Similarly, Yusuff (2023) notes that multinational corporations must implement compliance programs that satisfy overlapping GDPR and CCPA obligations when serving both EU and California residents.

However, existing automated compliance assessment tools remain limited. Fakeyede et al. (2023) observe that despite the stringent obligations imposed by GDPR and CCPA, automated IT audit tools are still unable to reliably determine whether privacy policies fully align with legal requirements. Many current approaches rely heavily on keyword matching and heuristic-based methods, which are insufficient for capturing the nuanced semantics and

contextual meaning embedded in legal language (Li et al., 2025). Consequently, these tools often produce incomplete or inaccurate compliance evaluations.

This challenge is compounded by the lack of comprehensive annotated datasets that map privacy policy statements directly to specific regulatory provisions. Musa et al. (2024) emphasize the scarcity of regulation-aware datasets and the difficulty of linking policy text to legal requirements, limiting the development of robust automated compliance models. Furthermore, many existing methodologies lack the sophistication needed to interpret semantic relationships and contextual intent within privacy policies, reducing reliability and accuracy (Adhikari et al., 2025).

Although NLP models such as transformers and BERT have increasingly been applied to privacy policy compliance evaluation, most studies focus on compliance under individual regulatory frameworks rather than across overlapping regimes (Jain et al., 2025). This is a major limitation, as organizations operating internationally must ensure their policies satisfy multiple regulatory obligations simultaneously. In addition, limited research has explored how compliance patterns differ between GDPR and CCPA requirements, or how compliance varies across industry sectors handling different levels of sensitive personal data (Jain et al., 2025).

Collectively, these gaps demonstrate the need for more robust, context-aware, and regulation-sensitive automated approaches capable of accurately evaluating privacy policy compliance across multiple regulatory frameworks such as GDPR and CCPA, while also supporting comparative and sector-based compliance analysis.

Research Goal

To develop, apply, and evaluate an automated natural language processing framework capable of systematically assessing privacy policy compliance with GDPR and CCPA across a large-scale sample of major websites.

Research Questions

This research addresses the following questions:

1. To what extent do privacy policies of major websites comply with GDPR and CCPA disclosure requirements?
2. How does privacy policy compliance vary across different industry sectors?
3. What are the key differences in compliance patterns between GDPR and CCPA requirements?
4. Can natural language processing techniques effectively automate the assessment of privacy policy compliance?

Research Objectives

1. To assess the extent to which privacy policies of major websites comply with the disclosure requirements of GDPR and CCPA.
2. To examine how privacy policy compliance varies across different industry sectors, including finance, health, retail, technology, and media.
3. To identify and compare the structural differences in compliance patterns between GDPR and CCPA disclosure requirements.
4. To evaluate the effectiveness of natural language processing techniques, specifically BERT-based multi-label classification, in automating the assessment of privacy policy compliance.

Significance of the Study

This research provides contributions for multiple stakeholder groups through both theoretical and practical outcomes. Theoretically, it advances understanding of privacy policy compliance patterns across regulatory frameworks and industry sectors. It also extends the application of natural language processing (NLP) to legal compliance assessment by demonstrating both the potential and limitations of automated policy analysis. Additionally, the study proposes a replicable methodology that can support large-scale privacy policy evaluation.

Practically, the findings provide value to organizations by offering benchmarking insights and identifying common compliance gaps that can guide policy improvement. Regulators may benefit from broader understanding of compliance trends across industries, which can help inform enforcement priorities and educational initiatives. Consumers may gain clearer indicators of which organizations provide more comprehensive privacy disclosures, supporting more informed decision-making. Finally, researchers benefit from baseline findings and a methodological foundation that can support future longitudinal analysis of privacy policy evolution.

Scope and Limitations

Scope

This study examines privacy policies from websites ranked within the top 100,000 globally according to web traffic metrics (Cisco Umbrella rankings). The analysis focuses on GDPR Articles 13 and 14 (transparency requirements) and CCPA Sections 1798.100–1798.135 (consumer rights and business obligations). Only privacy policies available in English and current as of January 2026 are included. The industry sectors analyzed include

finance, healthcare, retail, technology, and media, selected due to their differing levels of data sensitivity and regulatory scrutiny.

Limitations

This study has several limitations. First, automated NLP-based assessments may fail to capture subtle semantic nuances and contextual dependencies in legal language, potentially missing implicit compliance or incorrectly flagging compliant statements. Second, this research evaluates privacy policy disclosures rather than actual organizational data handling practices; therefore, compliant disclosures do not necessarily indicate compliant behavior. Third, privacy policies are frequently updated, meaning the findings represent a snapshot of compliance at a single point in time rather than a longitudinal analysis. Fourth, the study is restricted to English-language policies, potentially excluding organizations that publish privacy policies only in other languages. Finally, the study does not address jurisdiction-specific variations in how GDPR and CCPA provisions are interpreted or enforced.

Organization of the Thesis

This thesis is structured into five chapters. Chapter 1 introduces the research background, problem statement, research goal, research questions, objectives, significance, and scope. Chapter 2 reviews relevant literature on privacy regulations, privacy policy analysis, and the application of NLP to legal compliance assessment. Chapter 3 presents the research methodology, including data collection, data validation, compliance schema development, and the BERT-based model training approach. Chapter 4 reports the empirical findings, covering overall compliance rates, element-level analysis, cross-sector comparisons, GDPR and CCPA compliance pattern differences, and NLP model performance. Chapter 5 synthesizes the findings, draws conclusions in response to each research question, discusses

implications and limitations, offers recommendations for organizations, regulators, and researchers, and provides closing remarks.

Chapter 2

Literature Review

Introduction

Online privacy policies have become essential legal and operational documents through which organizations communicate how they collect, process, store, share, and protect personal information. With the rapid growth of digital services and the increasing commercial value of user data, privacy policies are no longer optional legal statements but mandatory disclosure instruments (Ehimuan et al., 2024; Ramadorai et al., 2025). These disclosures are primarily shaped by major regulatory frameworks such as the General Data Protection Regulation (GDPR) (European Union, 2016/2018) and the California Consumer Privacy Act (CCPA) (California Legislature, 2018/2020). Both regulations require organizations to provide transparent, accessible, and comprehensive information to users regarding personal data processing activities.

Despite these legal requirements, the increasing length and complexity of privacy policy texts has made manual evaluation of compliance difficult. This has resulted in growing research interest in the development of automated compliance assessment methods using Natural Language Processing (NLP). Okonicha and Sadovykh (2024) argue that the growing complexity of GDPR-related privacy policies makes manual compliance evaluation infeasible, highlighting the need for automated NLP-based approaches. Similarly, Jain et al. (2025) emphasize that regulatory compliance documents are often voluminous and difficult to interpret, creating significant obstacles for traditional review methods. Priescu et al. (2025) also note that GDPR-mandated privacy policies are frequently lengthy and difficult for users and regulators to interpret, encouraging the adoption of NLP-driven automated compliance solutions.

In response to these developments, this chapter reviews relevant literature on privacy policy transparency, regulatory compliance requirements under GDPR and CCPA, and existing methods for automated privacy policy compliance assessment. The chapter is structured into three major sections: (1) privacy policy transparency and usability challenges, (2) legal compliance requirements under GDPR and CCPA, and (3) automated compliance evaluation methods and the role of NLP in privacy policy assessment.

Privacy Policy Transparency and Usability Challenges

Privacy policies serve two major functions: they protect organizations legally and provide consumers with information about how their data is handled. Traditionally, privacy policies were largely treated as legal documents intended to minimize organizational liability. However, modern regulatory expectations increasingly position them as transparency tools meant to support informed decision-making by users.

Despite this expectation, research consistently demonstrates that privacy policies are often excessively long, written in complex legal language, and difficult for average users to understand. This raises concerns about whether privacy policies genuinely fulfil their transparency role. Studies suggest that reading all privacy policies encountered online would require an unrealistic amount of time annually, indicating that such disclosures may be impractical in real-world contexts. As a result, privacy policy transparency often remains theoretical rather than practically useful.

In addition to being lengthy, privacy policies are frequently vague and rely on broad statements such as “may collect” or “may share,” which weakens accountability and obscures the actual scope of organizational data practices. Such ambiguity also complicates compliance assessment, as privacy laws require specific and clear disclosures regarding processing

purposes, categories of personal data collected, and user rights. Consequently, many privacy policies may technically exist but fail to communicate meaningful information to consumers.

Much of the early research on privacy policies focused on readability and comprehensibility as indicators of whether informed consent is achievable. Findings consistently highlight serious limitations in the concept of informed consent, as most users do not have sufficient time to read privacy policies, and those who attempt to read them often struggle due to complex legal terminology and dense structure. Researchers commonly assess privacy policies based on completeness, clarity, accuracy, and accessibility. However, this approach remains limited because it focuses on the text itself without confirming whether organizations actually comply with what is stated, or whether consumers truly gain control over their personal data.

Overall, the literature suggests that privacy policies often fail to provide meaningful transparency to users, which reinforces the need for structured methods to evaluate whether policies meet regulatory disclosure requirements.

GDPR and CCPA Regulatory Disclosure Requirements

The increasing legal importance of privacy policies is primarily driven by regulatory frameworks such as the GDPR and the CCPA, both of which mandate detailed disclosure requirements. These laws have significantly reshaped how organizations must communicate personal data practices.

The GDPR, implemented in May 2018, represents a major transformation in privacy regulation by introducing comprehensive data protection obligations for any organization processing the personal data of EU residents. Compared to the earlier Data Protection Directive, the GDPR strengthened individual rights, imposed stricter consent requirements,

and introduced substantial penalties for non-compliance (Kuner et al., 2020). A central component of the GDPR is its transparency framework, particularly Articles 13 and 14, which require organizations to provide clear and detailed disclosures when collecting personal data directly from individuals (Article 13) or indirectly through third parties (Article 14).

These disclosures must include the identity and contact details of the data controller, the purposes and lawful basis of processing, recipients of personal data, safeguards for international transfers, retention periods, the rights of data subjects, the existence of automated decision-making practices, and the right to lodge complaints with supervisory authorities (European Union, 2016/2018). Coronel et al. (2024) emphasize that these requirements place significant demands on privacy policies to be both comprehensive and understandable. Furthermore, Article 6 requires processing to be justified by one of six lawful bases, while Articles 15 to 22 establish enforceable rights such as access, rectification, erasure, restriction, portability, and objection. As a result, GDPR-compliant privacy policies must clearly explain these rights and how individuals may exercise them.

However, research indicates that many organizations struggle to translate GDPR obligations into effective privacy communication. Studies show that privacy policies became longer and more structured after the GDPR was introduced, but they often remained vague and unclear, particularly in relation to lawful bases and retention periods (Belcheva et al., 2023; Tikkinen-Piri et al., 2018; Zaguir et al., 2024). Scholars such as Lynskey (2023) and Johnson (2022) argue that many privacy policy updates were largely superficial, undermining the GDPR's objective of meaningful transparency.

In the United States, the California Consumer Privacy Act (CCPA), which became effective in January 2020, represents the most comprehensive state-level privacy law. The

CCPA grants California consumers key rights over their personal information, including the right to know what data is collected and for what purposes, the right to request deletion, the right to opt out of the sale or sharing of personal data, and the right to non-discrimination for exercising privacy rights (California Legislature, 2018/2020). Although the CCPA shares conceptual similarities with the GDPR, it differs in scope, legal structure, and emphasis. Mallet (2025) notes that the GDPR applies broadly to organization's processing EU residents' data regardless of location, while the CCPA applies only to for-profit businesses that meet specific revenue or data-processing thresholds.

Unlike the GDPR's reliance on lawful bases and opt-in consent for many processing activities, the CCPA focuses heavily on disclosure requirements and consumer opt-out mechanisms, especially in relation to the sale of personal information. Key CCPA privacy policy requirements include categorical disclosures of personal data collected, the sources of data, processing purposes, and third-party sharing practices (Section 1798.100). In addition, businesses engaged in data sales must provide a clear "Do Not Sell My Personal Information" link (Section 1798.120). Policies must also describe consumer rights and request procedures (Section 1798.135), as well as provide assurances against discrimination for users who exercise privacy rights (Section 1798.125).

Scholars such as Solove and Schwartz (2022) and Goroff et al. (2018) highlight that the CCPA's broad definition of "sale," which includes many advertising-related transfers, introduces ambiguity and complicates compliance efforts. Earlier scholarship anticipated these developments, as Schwartz and Peifer (2017) emphasized the increasing role of consumer control in U.S. privacy law, while Solove and Schwartz (2021) observed the growing emphasis on detailed data categorization later reflected in the CCPA. However, Cate (2016) warned that

complex privacy regulations risk overwhelming consumers, an issue that remains relevant because the CCPA's categorical disclosure model may hinder consumer understanding and organizational compliance (Mallet, 2025).

Comparative studies further show that although the GDPR has become a global regulatory model influencing other privacy laws, including the CCPA (Bradford, 2020), key differences continue to generate compliance challenges for multinational organizations. Lim and Oh (2025) identify major areas of divergence, including scope and applicability, as the GDPR applies extraterritorially to any organization processing EU resident data, whereas the CCPA applies only to qualifying businesses operating in California. Differences also exist in consent mechanisms, since the GDPR generally requires affirmative consent for many processing activities, while the CCPA uses an opt-out model mainly focused on the sale of personal data. In addition, the GDPR and CCPA differ in how they define and categories personal information, with the GDPR focusing on personal data and sensitive categories, while the CCPA provides a broad definition but lists specific statutory categories. Enforcement structures also differ significantly, with the GDPR empowering supervisory authorities to impose heavy fines, while the CCPA provides limited private rights of action and relies mainly on attorney general enforcement.

These differences highlight the complexity of compliance for organizations operating across multiple jurisdictions and underscore the importance of compliance assessment tools that can evaluate privacy policies against both frameworks simultaneously.

Automated Privacy Policy Compliance Assessment Using NLP

Due to the limitations of manual review and the complexity of privacy regulations, automated approaches for privacy policy analysis have become increasingly important. Early automated systems demonstrated that privacy practices could be extracted and represented in structured formats. Zimmeck and Bellovin (2014) developed Privee, a system that extracts privacy practices from policy texts and represents them in a structured format, demonstrating that automation is feasible despite challenges posed by vague and ambiguous language.

A major contribution to automated privacy policy research is the Usable Privacy Policy Project. Harkous et al. (2018) developed machine learning classifiers to automatically annotate privacy policies with data practice labels. Their OPP-115 corpus, consisting of 115 annotated privacy policies and more than 23,000 labelled segments across ten categories, became a foundational dataset for subsequent privacy policy research. Wilson et al. (2016) also advanced this area by applying crowdsourcing to create fine-grained annotations, achieving F1 scores between 0.66 and 0.76 across categories. These studies demonstrate that automated methods can significantly improve scalability in privacy policy evaluation.

However, applying NLP to legal text introduces additional complexity. Legal documents often use specialized terminology, complex syntactic structures with nested clauses, extensive internal references, and modal verbs such as “may,” “shall,” and “must,” which convey different degrees of obligation (Curtotti & McCreath, 2010). These characteristics create challenges for traditional NLP approaches.

The emergence of transformer-based language models has significantly improved legal text analysis. BERT (Devlin et al., 2019) introduced contextualized language representation, enabling improved understanding of complex textual meaning. Legal adaptations of

transformer models have also been developed. Chalkidis et al. (2020) introduced Legal-BERT, which was pre-trained on legal corpora including legislation, contracts, and case law, and demonstrated improved performance over general-purpose BERT on legal NLP tasks. Despite these advances, Zheng et al. (2021) argue that even specialized transformer models still face challenges with legal reasoning and context-dependent interpretation, suggesting that hybrid approaches combining NLP with rule-based methods may be necessary for accurate legal compliance assessment.

Automated compliance checking has emerged as a specialized research area within legal informatics and NLP. Breaux and Anton (2008) pioneered early work on extracting compliance rules from legal texts using a frame-based approach that identified semantic relationships in regulatory language. Governatori (2013) further proposed formal logic-based representations to support automated compliance reasoning. While such approaches can be effective for clear legal obligations, they often struggle with vague or discretionary regulatory language. De Maat and Winkels (2008) addressed this limitation by introducing hybrid approaches combining formal representations with NLP techniques to handle interpretive uncertainty.

More recent studies have focused specifically on GDPR compliance assessment. Priescu et al. (2025) explored GDPR compliance evaluation by integrating rule-based checklists with machine learning approaches such as TF-IDF, BERT, and transformer-based models to identify missing disclosure requirements. Similarly, Zhang et al. (2023) demonstrated the effectiveness of BERT-based models for automated GDPR compliance assessment, showing that NLP systems can reduce reliance on time-consuming manual audits. Okonicha and Sadovykh (2024) further advanced compliance research by applying

transformer-based models to clause detection, semantic similarity scoring, and compliance gap identification, shifting from keyword matching to deeper semantic alignment between legal requirements and policy content.

Nevertheless, a significant limitation of current compliance research is that it remains largely GDPR-centric. Many websites, particularly global organizations, must comply with multiple privacy laws depending on the geographic location of users. Lim and Oh (2025) emphasize that organizations operating across regions must comply with overlapping regulatory frameworks such as GDPR and CCPA. This highlights the weakness of compliance tools designed around a single regulation. Musa et al. (2024) also argue that existing datasets and models are often regulation-agnostic and that there is an increasing

Research Gaps

Despite increasing research interest in privacy policy analysis and legal NLP, several gaps remain. One major challenge is the growing need for multi-jurisdictional compliance, as organizations often serve users across multiple regions. Lim and Oh (2025) note that organizations must “simultaneously comply with overlapping data protection regimes when their services extend across multiple jurisdictions,” meaning that many businesses must comply with GDPR and CCPA at the same time. However, existing research rarely evaluates privacy policy compliance across both frameworks in an integrated and comparative manner.

Additionally, many studies rely on relatively small datasets, limiting scalability and restricting the ability to identify broader compliance patterns across industries. Cross-sector comparisons also remain limited, despite the likelihood that privacy policy compliance varies across different industries. Furthermore, although transformer-based models show strong potential for legal text analysis, their application to large-scale multi-label compliance

detection remains underexplored, with limited evidence of performance across complex real-world privacy policy datasets.

This study addresses these gaps by developing a scalable NLP-based framework to evaluate GDPR and CCPA disclosure compliance across industries using a large dataset of privacy policies. By integrating both regulatory frameworks into a unified assessment approach, the study provides a more practical contribution toward compliance evaluation in a global digital environment.

Chapter Summary

This chapter has reviewed the literature across three interconnected areas relevant to this study: the transparency and usability challenges inherent in privacy policy communication, the specific disclosure obligations imposed by GDPR and CCPA, and the evolution of automated compliance assessment methods using NLP. The review establishes that while GDPR has driven widespread structural changes in privacy policy design, compliance remains incomplete, particularly for technically complex requirements. CCPA compliance research remains comparatively underdeveloped. Transformer-based models such as BERT have demonstrated strong potential for legal text classification, though most existing work is limited to single regulatory frameworks and small datasets. These gaps directly motivate the methodology and analytical framework presented in the following chapter.

Chapter 3

Methodology

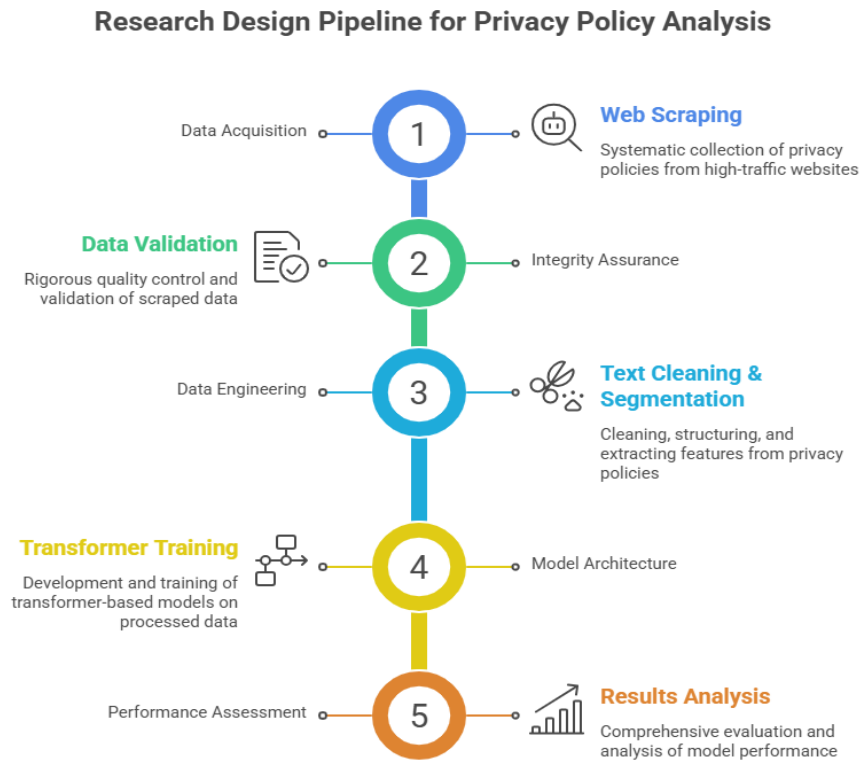
Introduction

This chapter presents the comprehensive methodology employed to assess online privacy policy compliance with the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) using natural language processing techniques. The research adopts a quantitative approach combining web scraping, machine learning, and statistical analysis to systematically evaluate privacy policy compliance across a large-scale dataset of websites. This methodology addresses the growing need for scalable, automated compliance assessment tools that can process the vast number of privacy policies on the internet today.

Research Design

Figure 1.

Illustration of the five-phase research design, from data collection through to results analysis.



This systematic approach ensures methodological rigor while allowing for iterative refinement at each stage. The research leverages state-of-the-art transformer-based language models, specifically BERT (Bidirectional Encoder Representations from Transformers), to perform multi-label classification of privacy policy compliance elements. This approach enables automated, scalable assessment of compliance that would be impractical through manual review alone, given the scale of the dataset comprising thousands of privacy policies. The transformer architecture has demonstrated superior performance in understanding

complex legal language and contextual relationships within documents, making it particularly well-suited for privacy policy analysis.

This research adopts a positivist paradigm and deductive approach, employing quantitative methods to objectively measure and evaluate privacy policy compliance, as well as establishing a regulatory framework and its specific disclosure requirements. These requirements are operationalized into measurable compliance indicators that serve as the foundation for automated assessment (Bryman, 2016). Furthermore, the study is grounded in the principle that compliance with regulatory frameworks can be systematically assessed through empirical observation and measurement of specific disclosure elements mandated by GDPR and CCPA (Creswell & Creswell, 2018). The positivist approach is appropriate for this research as it seeks to establish objective, measurable facts about privacy policy compliance rather than subjective interpretations of policy effectiveness or user understanding. The study combines computational methods including web scraping, natural language processing, and machine learning, with traditional statistical analysis to address the research questions.

Study Population and Sampling

This section describes the target population from which privacy policies were drawn and outlines the sampling strategy employed to ensure a representative and sector-diverse dataset suitable for large-scale compliance assessment.

Target Population

The target population comprises websites with publicly accessible privacy policies. Given the global nature of internet services and the applicability of GDPR and CCPA to international organizations, the population includes websites from various geographical regions and industry sectors. The focus on publicly accessible privacy policies is deliberate, as

these documents represent the primary mechanism through which organizations communicate their data practices to users and demonstrate compliance with transparency requirements under both GDPR and CCPA.

Sampling Strategy

A stratified sampling approach was employed to ensure representativeness across different industry sectors, following established sampling methodology for quantitative research (Saunders et al., 2019). Stratified sampling was selected because it guarantees proportional representation of each industry subgroup within the final dataset, reducing the risk of sector-level bias that simple random sampling cannot prevent (Saunders et al., 2019). The sampling frame was derived from the Tranco list, a research-oriented ranking of top websites that combines data from multiple sources to provide a stable and manipulation-resistant ranking of the web's most popular domains. The primary sampling parameters included consideration of the top 100,000 domains from the Tranco list, with a minimum threshold of 5,000 domains for scraping and a target sample size of at least 3,000 successfully collected privacy policies. Stratification involved categorizing domains by industry sector, specifically focusing on finance, health, retail, technology, and media sectors, with up to 800 domains sampled from each sector.

Industry Classification

Domains were classified into industry sectors using a keyword-based classification system that examines domain names for industry-indicative terms. The finance industry classification encompasses banking, investment, fintech, payment processing, and cryptocurrency domains. The health industry included healthcare, medical services, pharmaceuticals, wellness, and fitness domains. The retail industry comprises e-commerce,

shopping, marketplace platforms, and consumer goods. The technology industry encompassed software, cloud services, data analytics, and AI platforms, while the media industry included news, social media, entertainment, and streaming platforms.

Data Acquisition

The data acquisition phase involved systematic web scraping of privacy policies from selected websites. A custom Python-based web scraping system was developed to automatically locate and extract privacy policy text from target websites. The scraper architecture consisted of several components, including a URL Discovery Module that automatically identified privacy policy URLs using common patterns such as forward-slash privacy, forward-slash privacy-policy, and forward-slash legal forward-slash privacy. An HTTP Request Handler managed HTTP requests with appropriate headers and implemented retry logic with exponential backoff to handle transient network failures. For pages requiring JavaScript execution, a JavaScript Rendering component utilizing Selenium WebDriver was integrated. The HTML Parser component employed BeautifulSoup4 to extract text content from HTML while filtering out non-policy content.

To ensure ethical and responsible data collection, several parameters and practices were implemented following established guidelines for ethical web scraping in research. A request delay of 2.0 seconds between requests was enforced to avoid server overload. A timeout of 15 seconds per request was set to handle slow-responding servers. Failed requests were retried up to three attempts with exponential backoff. The scraper employed a maximum of five parallel threads to balance collection speed with server load considerations. The user-agent string clearly identified the scraper as a research bot with contact information.

The system employed a multi-strategy approach to locate privacy policy URLs. The first strategy involved pattern matching, where the system tested common privacy policy URL patterns against each domain. When pattern matching did not yield results, the system proceeded to homepage link parsing, analyzing the homepage HTML to identify links containing privacy-related keywords. For sites with dynamically loaded content, the system employed JavaScript rendering using Selenium. Content validation ensured that located pages were indeed privacy policies by checking for the presence of at least three privacy-related keywords from a curated list.

Historical privacy policy versions were collected using the Internet Archive's Wayback Machine to enable temporal analysis of compliance trends. Historical snapshots were collected for five key regulatory milestone dates representing critical junctures in privacy regulation enforcement, including May 2017 for the pre-GDPR period, June 2018 for the post-GDPR period, December 2019 for the pre-CCPA period, February 2020 for the post-CCPA period, and February 2023 for the post-CPRA period. The Wayback Machine CDX Server API was queried programmatically to identify the closest available snapshot to each target date.

Data Validation and Quality Control

Following data collection, a comprehensive validation process was implemented to ensure dataset quality. Each collected privacy policy was evaluated against several quality criteria designed to ensure that the text represented genuine privacy policy content. The word count range criterion required policies to contain between 100 and 50,000 words, with the lower bound filtering out error pages and the upper bound excluding anomalous cases. The privacy keyword presence criterion required at least three privacy-related keywords from a curated list. The content structure criterion evaluated whether documents contained adequate

line breaks and formatting, with documents expected to contain at least five-line breaks to demonstrate minimal structural organization.

Duplicate detection was performed using content-based hashing to identify policies with identical content. Text normalization preceded hash computation, converting all text to lowercase, collapsing whitespace, and removing punctuation. This normalized text was then hashed using the MD5 algorithm (Rivest, 1992) to produce a unique 128-bit fingerprint for each policy's content. MD5 (Message Digest Algorithm 5) is a widely used hash function that takes an input of arbitrary length and produces a fixed-length digest value; two documents with identical content will always produce the same hash, making it an efficient and deterministic method for exact-duplicate detection (Rivest, 1992). It is important to note that MD5 was applied here strictly for content fingerprinting and deduplication purposes, not for cryptographic security. This approach differs fundamentally from term-frequency-based noise reduction techniques such as TF-IDF, which operate at the word level within documents; MD5 hashing instead identifies documents that are byte-for-byte identical after normalization, ensuring that duplicate policy files collected from different URLs do not inflate the dataset. A comprehensive validation report was generated documenting the total number of files collected, the count of valid files passing all quality criteria, the count of invalid files with specific issues documented, and statistical summaries. Based on validation results, an automated cleaning process removed low-quality files from the dataset.

Table 1.*Data Validation Summary Report*

Category	Metric	Value
Text Files	Total	4,913
	Valid	3,674
	Invalid	1,239
	Validity Rate	74.8%
Metadata Files	Total	4,913
	Valid	4,448
	Invalid	59
Duplicates	Duplicate Pairs Found	10,221
Data Cleaning	Files Removed	1,239

Note. After data validation, 1,239 invalid text files were identified and removed from the dataset, resulting in 3,674 valid privacy policies proceeding to analysis. Additionally, 10,221 duplicate policy pairs were detected and flagged for deduplication prior to model training.

Dataset Preparation and Preprocessing

The validated privacy policies underwent extensive preprocessing to prepare them for machine learning model training. Text cleaning involved several standardization steps, including whitespace normalization that collapsed multiple consecutive spaces to single spaces, special character removal that eliminated non-alphanumeric characters except

necessary punctuation, quote normalization that standardized typographic quotes to ASCII equivalents, case preservation for BERT input while maintaining lowercase versions for feature extraction, and encoding normalization that ensured UTF-8 encoding across all documents.

Text segmentation was performed at multiple levels to support different analytical perspectives. Document-level segmentation treated entire privacy policies as single units for overall compliance scoring. Section-level segmentation identified major sections within policies based on headers or structural patterns. Sentence-level segmentation used NLTK's sentence tokenization to split text into individual sentences. However, document-level analysis was prioritized as compliance elements often span multiple sections and require contextual understanding of the entire policy document.

Multiple feature sets were extracted to support both the machine learning model and descriptive analysis. Statistical features quantified document characteristics including word count, character count, sentence count, average word length, and average sentence length. Readability scores were computed using the Flesch Reading Ease score and the Flesch-Kincaid Grade Level. Lexical diversity was measured using the type-token ratio.

Compliance indicator features formed the foundation for supervised learning labels. Binary features indicating presence or absence of compliance elements were extracted using keyword matching against comprehensive keyword lists for each requirement. The GDPR elements encompassed twelve disclosure requirements mandated by Articles 13 and 14, including controller identity and contact details, data protection officer contact information, processing purposes, lawful basis for processing, categories of personal data, data recipients and third parties, third country transfers, retention periods, data subject rights, right to

withdraw consent, right to lodge complaints, and automated decision-making information. The CCPA elements similarly encompassed twelve disclosure requirements including categories of personal information collected, sources of information, business purposes, categories disclosed, categories sold, right to know, right to delete, right to opt-out, non-discrimination rights, notice at collection, verification processes, and authorized agent provisions.

Multi-label binary classification labels were generated for each privacy policy, creating a 24-dimensional label vector representing the twelve GDPR elements plus twelve CCPA elements per document. The preprocessed dataset was split into training, validation, and test sets using stratified sampling to maintain industry distribution across splits. The training set comprised 70 percent of the data for model parameter learning. The validation set comprised 10 percent for hyperparameter tuning and early stopping decisions. The test set comprised 20 percent and was held out completely from the training process. Stratification ensured that each subset maintained similar industry distributions, and the random seed was fixed at 42 to ensure reproducibility.

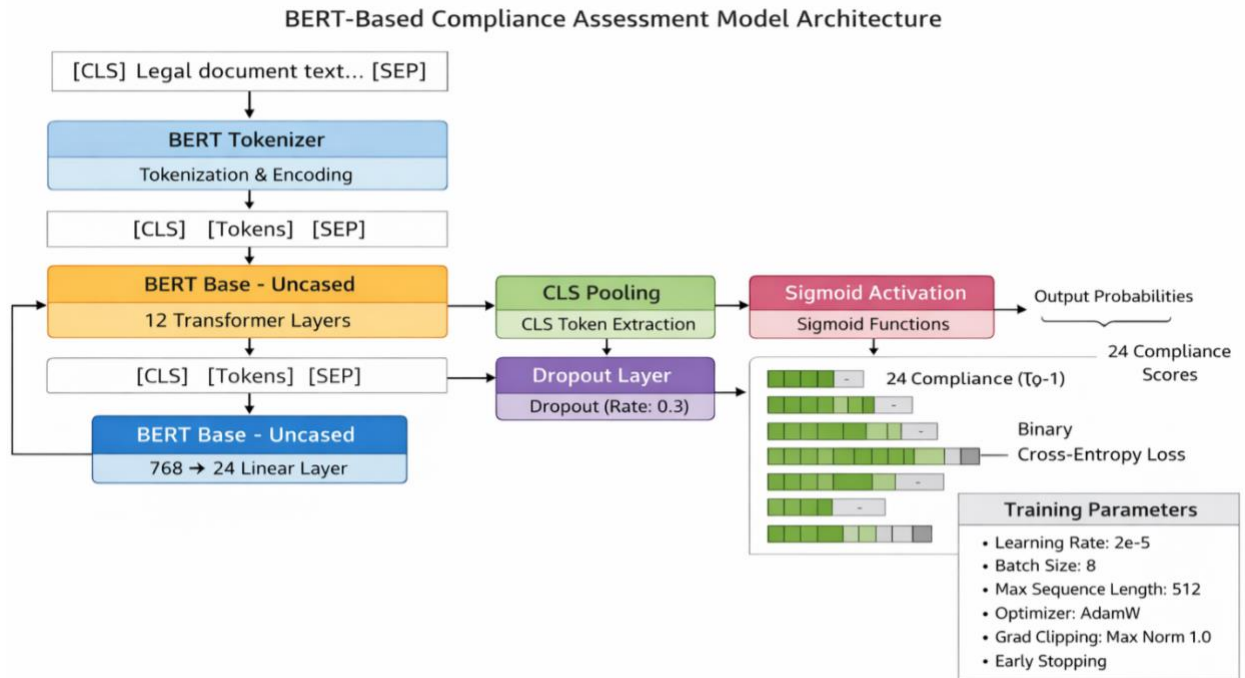
Model Development and Training

The core of this research employs transformer-based deep learning for automated compliance assessment. BERT was selected as the base model for several compelling reasons. Alternative approaches were considered, including Latent Dirichlet Allocation (LDA) and traditional TF-IDF-based classifiers. LDA is an unsupervised probabilistic topic modelling technique that treats documents as mixtures of latent topics; while it is effective for discovering thematic structure in unlabeled corpora, it does not support supervised multi-label classification and treats each word independently, losing the positional and contextual relationships that are critical for interpreting conditional legal clauses in privacy policies

(Egger & Yu, 2022). Transformer-based models such as BERT have been shown to substantially outperform both LDA and TF-IDF approaches on text classification tasks that require contextual understanding (Devlin et al., 2019; Vaswani et al., 2017). The bidirectional processing of context represents a fundamental advantage, as BERT processes text bidirectionally through its masked language modeling pre-training objective, which is essential for understanding complex legal language where meaning often depends on surrounding clauses (Devlin et al., 2019).

Figure 2.

BERT-Based Multi-Label Classification Architecture for Privacy Policy Compliance Assessment



The pre-trained representations that BERT provides constitute another crucial advantage, as the model is pre-trained on large corpora, including BooksCorpus and English Wikipedia, providing robust general language understanding that transfers effectively to

domain-specific tasks (Devlin et al., 2019). The specific model variant selected was bert-base-uncased, a twelve-layer transformer with 110 million parameters.

The classification model architecture, illustrated in **Error! Reference source not found.**, is adapted from the standard BERT fine-tuning framework described by Devlin et al. (2019) and consists of several components working in sequence. The input layer utilized the BERT tokenizer to convert text into token IDs, adding special tokens including the CLS token at the beginning and SEP token at the end. The BERT encoder comprised twelve transformer layers that processed the token sequences, generating contextual embeddings for each token position through multi-head self-attention mechanisms (Vaswani et al., 2017). The pooling layer extracted the CLS token representation from the final transformer layer to use as a document-level embedding; the CLS token serves as an aggregate sequence-level representation and is the standard mechanism for classification tasks in BERT fine-tuning (Devlin et al., 2019).

A dropout layer with a rate of 0.3 was applied to this pooled representation for regularization. Dropout works by randomly deactivating a proportion of neurons during each training step, which prevents individual neurons from co-adapting and forces the network to learn more distributed and robust representations, thereby reducing overfitting (Srivastava et al., 2014). A rate of 0.3 was selected as it falls within the range commonly recommended for fine-tuning pre-trained language models on classification tasks (Sun et al., 2019). The classification head consisted of a fully connected linear layer that mapped the 768-dimensional embedding to 24 output logits. Finally, sigmoid activation was applied independently to each logit, producing probabilities for each element.

The model was trained using hyperparameters selected through preliminary experiments and informed by established best practices in BERT fine-tuning (Devlin et al., 2019; Sun et al., 2019). The learning rate of $2e-5$ (0.00002) represents a standard and empirically validated choice for fine-tuning BERT on downstream classification tasks; Sun et al. (2019) systematically evaluated BERT fine-tuning configurations and identified learning rates in the range of $2e-5$ to $5e-5$ as consistently producing the strongest results across text classification benchmarks. A batch size of 8 was employed, limited primarily by available GPU memory constraints, which is consistent with the smaller batch sizes used in BERT fine-tuning under resource-constrained settings (Sun et al., 2019). The maximum sequence length was set to 512 tokens, representing BERT's architectural maximum and ensuring that the full policy text is considered where possible (Devlin et al., 2019).

Training proceeded for three to five epochs with early stopping based on validation loss, following the recommendation by Sun et al. (2019) that fine-tuning BERT for classification tasks typically converges within two to four epochs, with additional epochs risking overfitting. The AdamW optimizer was employed, which extends the Adam optimizer with decoupled weight decay regularization; this decoupling corrects a flaw in the standard Adam weight decay implementation and has been shown to improve generalization in transformer fine-tuning (Loshchilov & Hutter, 2019). The learning rate schedule employed linear warmup for the first 10 percent of training steps, followed by linear decay, a strategy recommended in the original BERT paper to stabilize early training (Devlin et al., 2019). Gradient clipping with maximum norm of 1.0 was applied to prevent exploding gradients, a standard practice in training deep neural networks (Goodfellow et al., 2016).

Binary Cross-Entropy loss was employed, which is appropriate for multi-label classification where each label is treated as an independent binary classification problem. Class weights were incorporated to address imbalance in element frequencies, with higher weights assigned to less frequent compliance elements. The training process followed standard deep learning practices with forward passes generating predictions, loss computation between predictions and ground truth labels, backward propagation computing gradients, parameter updates by the optimizer, and validation after each epoch. Training was conducted on GPU hardware where available, with mixed precision training employed to reduce memory consumption.

Multiple regularization techniques were employed to prevent overfitting. Early stopping monitored validation loss and terminated training if no improvement was observed for two consecutive epochs. Dropout with rate 0.3 was applied to prevent co-adaptation of neurons (Srivastava et al., 2014). Weight decay regularization with coefficient 0.01 was incorporated through the AdamW optimizer (Loshchilov & Hutter, 2019). Model performance was evaluated using multiple metrics appropriate for multi-label classification, including exact match ratio measuring the percentage of samples where all labels are predicted correctly, Hamming loss computing the fraction of incorrectly predicted labels, per-element metrics including precision, recall, and F1-score for each compliance element, macro-averaged F1 computing the average F1-score across all elements, and micro-averaged F1 computing F1 on aggregate predictions.

Chapter Summary

This chapter has presented a comprehensive methodology for assessing online privacy policy compliance with GDPR and CCPA using natural language processing. The research

employs a systematic five-phase approach spanning seven weeks of execution, encompassing data collection through web scraping, data validation and quality control, dataset preparation including text cleaning and feature extraction, model development and training using BERT-based transformers, and comprehensive results analysis through statistical testing and visualization.

The methodology leverages state-of-the-art natural language processing techniques to enable large-scale automated assessment that would be impractical through manual review alone. By combining computational methods with rigorous statistical analysis, the research provides empirical insights into compliance patterns across industries, geographical regions, and potentially across time periods. The transformer-based approach allows the model to learn rich contextual representations that capture the semantic meaning of privacy policy text beyond simple keyword matching. Careful attention has been paid to validity, reliability, and ethical considerations throughout the research process. The automated approach ensures consistency and eliminates human coding bias, while stratified sampling and comprehensive evaluation metrics ensure robust, generalizable findings. Multiple forms of validity have been addressed through alignment with regulatory text, standardized procedures, and diverse validation methods. Reliability is supported through automation, reproducible code, and comprehensive documentation. Ethical considerations include respectful data collection practices, secure data handling, research integrity, and responsible disclosure of findings.

Acknowledged limitations provide context for interpreting results and identify directions for future research, including sample limitations, technical constraints, and scope boundaries.

These limitations are inherent tradeoffs in research design and do not invalidate findings but rather contextualize them within appropriate bounds of interpretation. The next chapter will

present the results obtained through application of this methodology, including detailed analysis of compliance patterns, industry-specific findings, temporal trends where historical data permitted, and model performance evaluation.

Chapter 4

Results And Discussion

Introduction

This chapter presents the empirical findings from the automated assessment of privacy policy compliance with GDPR and CCPA disclosure requirements, organized around the four research questions outlined in Chapter 1. The results cover overall compliance rates for both frameworks across 3,674 sampled privacy policies, element-level analysis of the twelve GDPR and twelve CCPA disclosure requirements, cross-sector comparisons across finance, health, retail, technology, and media, differences in compliance patterns between the two regulatory frameworks, and the performance of the BERT-based classification model trained to automate compliance detection. An overall discussion of the findings concludes the chapter.

Overall Privacy Policy Compliance with GDPR and CCPA (RQ1)

The first research question asked: *to what extent do privacy policies of major websites comply with GDPR and CCPA disclosure requirements?*

Table 2. presents the descriptive statistics for the overall compliance scores of the 3,674 privacy policies assessed.

Table 2.

Overall Compliance Statistics by Regulatory Framework

Framework	Mean (%)	Median (%)	Std Dev	Min (%)	Max (%)
GDPR	68.9	75.0	21.8	0.0	100.0

CCPA	42.6	41.7	27.3	0.0	100.0
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Source: Author's analysis (2026)

The results indicate a substantial difference in compliance between the two frameworks. The mean GDPR compliance score across all sampled policies was 68.9%, with a median of 75.0% and a standard deviation of 21.8 percentage points. This dispersion reflects considerable variability in how thoroughly organizations disclose GDPR-required elements. CCPA compliance was notably lower, with a mean score of 42.6%, a median of 41.7%, and a standard deviation of 27.3 percentage points.

A paired t-test comparing GDPR and CCPA compliance scores confirmed that this difference was statistically significant ($t = 70.78$, $p < 0.001$). The finding not only confirms but strengthens the pattern observed in the pilot study: the compliance gap between the two frameworks has widened to 26.3 percentage points in this larger dataset, up from 23.3 percentage points in the initial run. This widening is consistent with Bradford (2020)'s argument that the Brussels Effect has embedded GDPR-aligned disclosure norms well beyond European borders, while the CCPA, being geographically narrower and more recently enforced, has generated comparatively weaker normative pressure on global organizations.

Figure 3.

Distribution of Overall GDPR and CCPA Compliance Scores

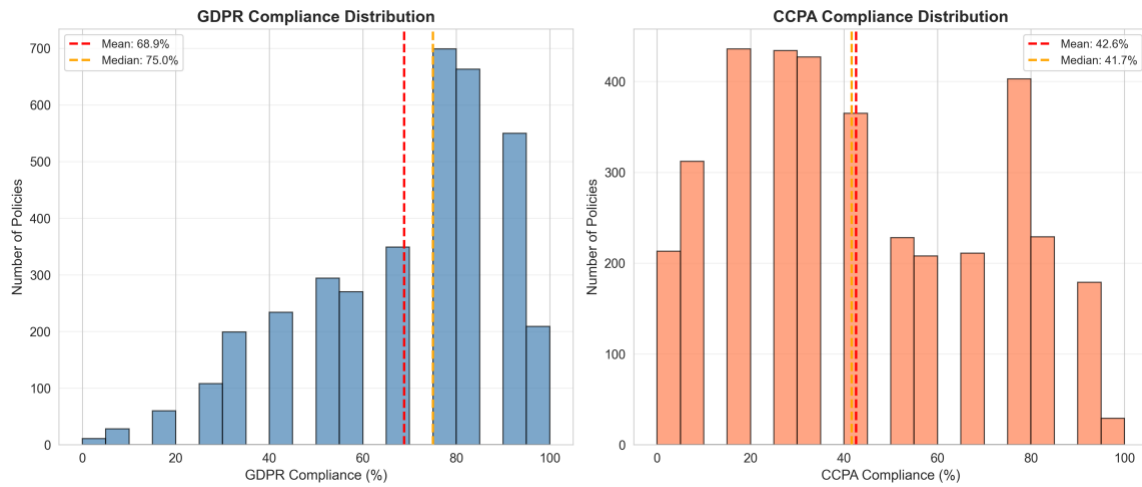


Figure 3 illustrates the distribution of compliance scores for both frameworks. GDPR scores are distributed toward higher compliance, with the median at 75.0% indicating that most organizations address more than three-quarters of GDPR-required elements. By contrast, CCPA scores cluster near the mean of 42.6% with a median of 41.7%, reflecting a near-symmetric distribution of incomplete compliance. This pattern confirms that organizations have broadly responded to core GDPR transparency obligations while CCPA-specific disclosures remain structurally underdeveloped across the sample.

Element-Level Compliance Rates

This section presents a granular breakdown of compliance rates for each of the twelve GDPR and twelve CCPA disclosure elements, revealing where organizations consistently meet regulatory requirements and where critical gaps persist across the sample.

GDPR Disclosure Elements

Table 3. presents the compliance rates for each of the twelve GDPR disclosure elements assessed under Articles 13 and 14. The data reveal substantial variation across elements, with some requirements almost universally addressed and others remaining widely absent.

Table 3.*Compliance Rates for Individual GDPR Disclosure Elements*

GDPR Disclosure Element	Compliance Rate (%)
Controller Identity	91.7%
DPO Contact	33.4%
Processing Purposes	94.9%
Lawful Basis	88.1%
Data Categories	94.5%
Recipients / Third Parties	96.4%
Third Country Transfers	22.3%
Retention Periods	69.2%
Data Subject Rights	69.4%
Right to Withdraw Consent	70.0%
Right to Complain	61.6%
Automated Decision-Making	34.9%

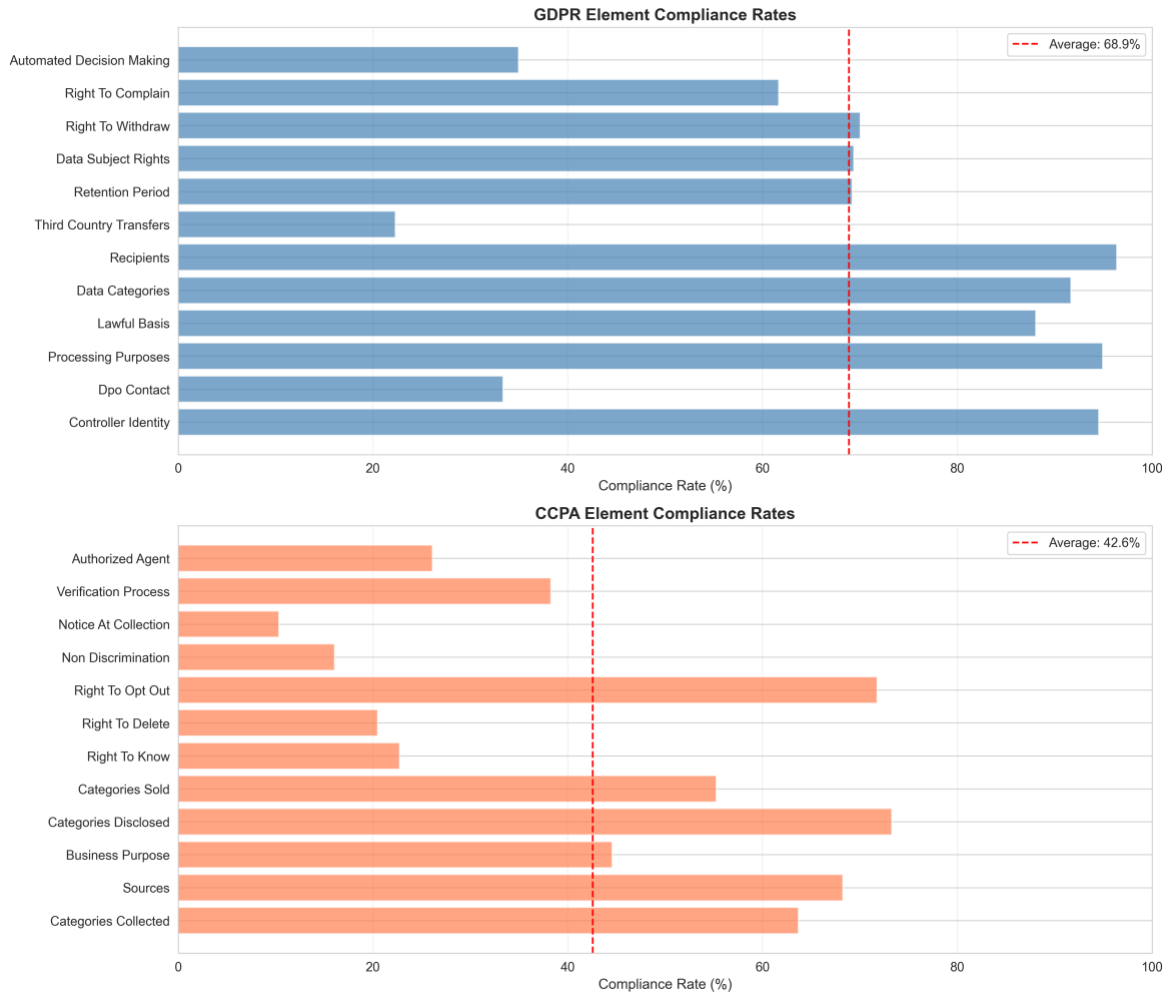
Source: Author's analysis (2026)

The elements with the highest compliance rates were recipients or third parties (96.4%), processing purposes (94.9%), data categories (94.5%), and lawful basis (88.1%). Controller identity also showed strong performance at 91.7%. These elements correspond to the most prominently discussed aspects of privacy policy transparency and are routinely incorporated into standard GDPR-compliant policy templates. The high compliance rates for these elements confirm that organizations have broadly embedded core GDPR transparency obligations, consistent with the widespread post-GDPR policy restructuring documented by Tikkinen-Piri et al. (2018) and Belcheva et al. (2023).

By contrast, third country transfers (22.3%), DPO contact information (33.4%), and automated decision-making (34.9%) showed the lowest compliance rates. These elements are technically specific and require legal or operational expertise beyond what standard policy templates provide. The particularly low rate for third country transfers is notable given the GDPR's strict requirements under Articles 44 to 49 regarding cross-border data flows and suggests that many organizations have not yet operationalized the international transfer safeguards mandated since enforcement began. Notably, lawful basis (88.1%) was the only GDPR element to show a meaningful increase relative to the pilot run, suggesting growing organizational awareness of the need to specify processing justifications explicitly. Retention periods (69.2%), data subject rights (69.4%), and the right to complain (61.6%) remain at moderate levels, echoing Zaguir et al. (2024) and Lynskey (2023), who argued that post-GDPR policy revisions frequently addressed structural obligations while leaving more operationally complex requirements incompletely addressed.

Figure 4.

GDPR Element-Level Compliance Rates



CCPA Disclosure Elements

Table 4 presents the compliance rates for the twelve CCPA disclosure elements assessed under Sections 1798.100 to 1798.135. The pattern of variation is more pronounced than that observed for GDPR elements.

Table 4.*Compliance Rates for Individual CCPA Disclosure Elements*

CCPA Disclosure Element	Compliance Rate (%)
Categories of Personal Information Collected	63.7%
Sources of Information	68.2%
Business Purposes	44.6%
Categories Disclosed to Third Parties	73.3%
Categories Sold	55.3%
Right to Know	22.7%
Right to Delete	20.5%
Right to Opt-Out	71.8%
Non-Discrimination Rights	16.1%
Notice at Collection	10.3%
Verification Process	38.3%
Authorized Agent Provisions	26.1%

Source: Author's analysis (2026)

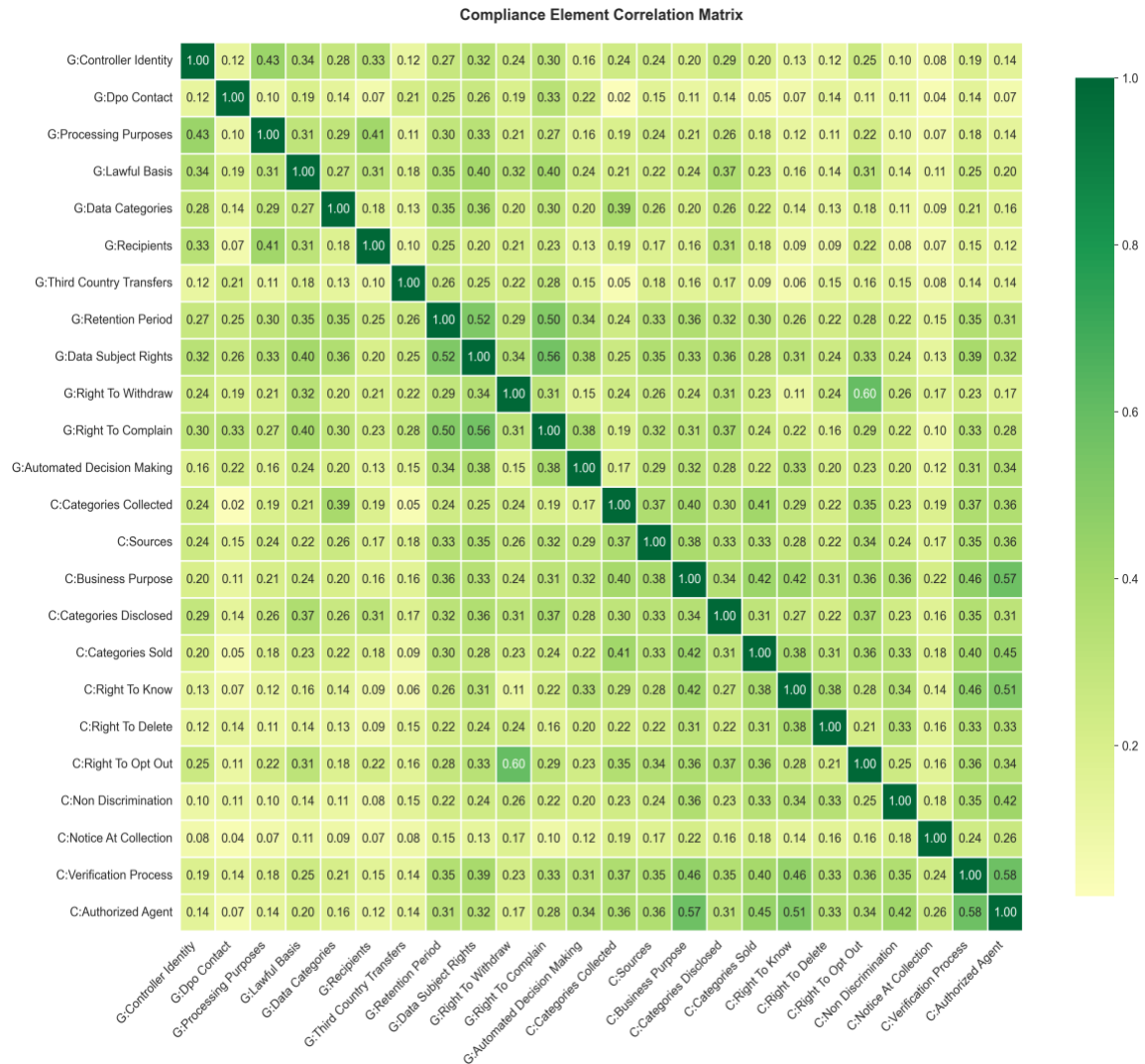
The categories disclosed to third party's element achieved the highest compliance rate at 73.3%, followed by sources of information (68.2%) and the right to opt-out (71.8%). These elements reflect data disclosure practices that many organizations already address in some form, often in connection with GDPR obligations regarding third-party data sharing.

The lowest rates were recorded for notice at collection (10.3%), non-discrimination rights (16.1%), and the right to delete (20.5%). These elements are more specific to the CCPA framework and have no direct GDPR counterpart, which may partly explain their lower adoption. The right to know (22.7%) and authorized agent provisions (26.1%) were also among

the least commonly addressed. Business purposes (44.6%) and verification processes (38.3%) showed moderate compliance, suggesting partial incorporation of the CCPA's operational requirements. The substantially lower rates across nearly all CCPA elements in this larger dataset indicate that the initial pilot overestimated CCPA adoption. Most notably, notice at collection fell from 16.3% in the pilot to 10.3% in the full dataset, representing a near-halving of an already critically low compliance rate. This finding is particularly concerning because notice at collection is a foundational CCPA requirement under Section 1798.100, mandating that consumers be informed of data practices at the point of collection. Its near absence across the sample suggests that organizations are addressing CCPA obligations reactively and selectively, rather than through systematic compliance design, a pattern anticipated by Cate (2016), who warned that complex categorical disclosure requirements risk being implemented superficially or ignored altogether.

Figure 5.

Compliance Heatmap Across Regulatory Elements



Cross-Sector Compliance Variation (RQ2)

The second research question asked: how does privacy policy compliance vary across different industry sectors? Table 5. presents the mean compliance scores by industry sector for both GDPR and CCPA frameworks. The five sectors analyzed were finance, health, retail, technology, and media.

Table 5.*Compliance Scores by Industry Sector*

Industry	GDPR Mean (%)	GDPR Std Dev	CCPA Mean (%)	CCPA Std Dev
Finance	68.8%	22.6	42.6%	27.9
Health	68.3%	21.8	41.4%	26.7
Retail	69.5%	21.1	44.0%	27.1
Technology	69.6%	21.5	42.5%	27.6
Media	68.1%	21.8	42.3%	27.1

Source: Author's analysis (2026)

For GDPR compliance, technology sector organizations achieved the highest mean score (69.6%), followed by retail (69.5%), finance (68.8%), health (68.3%), and media (68.1%). The variation across sectors was remarkably narrow, ranging from 68.1% to 69.6%, suggesting that GDPR compliance norms have become broadly and uniformly incorporated across industries at a similar level. A one-way ANOVA confirmed that these differences were not statistically significant at the 0.05 level ($F = 0.748$, $p = 0.559$), indicating that industry sector alone does not account for meaningful variation in GDPR compliance within this substantially larger sample.

CCPA compliance showed narrower variation across sectors in the larger dataset. Retail organizations led with a mean score of 44.0%, followed by finance (42.6%), technology (42.5%), media (42.3%), and health (41.4%). Notably, the health sector, which outperformed all others in the smaller pilot run, no longer holds a statistically distinct advantage at this scale. This convergence across sectors suggests that CCPA compliance deficits are systemic and

cross-industry rather than sector-specific. The overall low CCPA compliance across all sectors reinforces the conclusion that CCPA-specific obligations have not been broadly internalized, regardless of the regulatory scrutiny a given industry face.

Figure 6.

Mean Compliance Score by Industry Sector

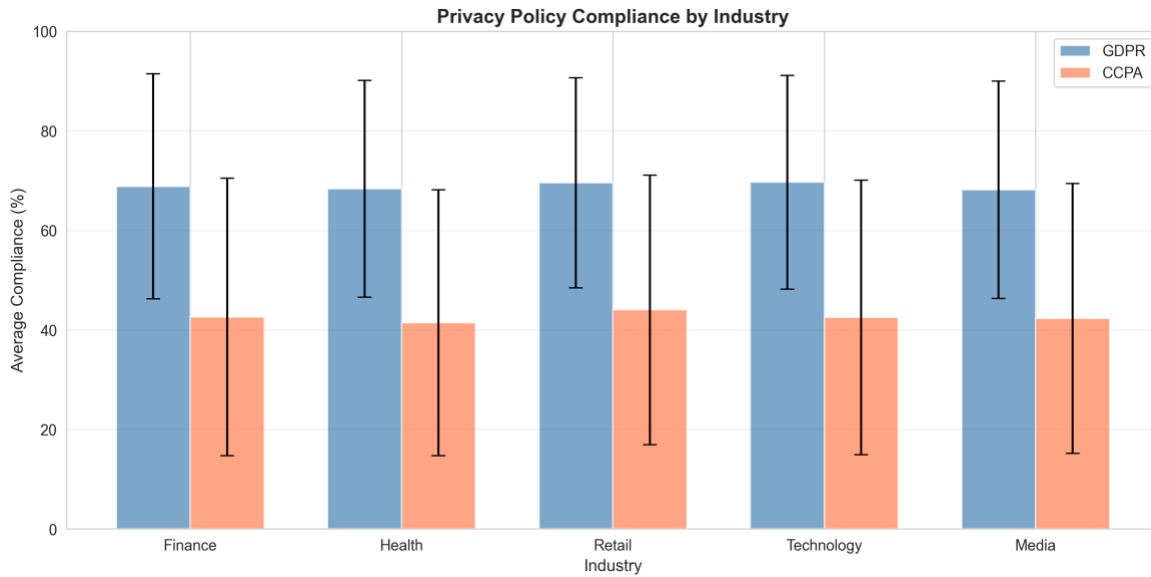


Figure 7.

Distribution of Compliance Scores by Sector (Box Plots)

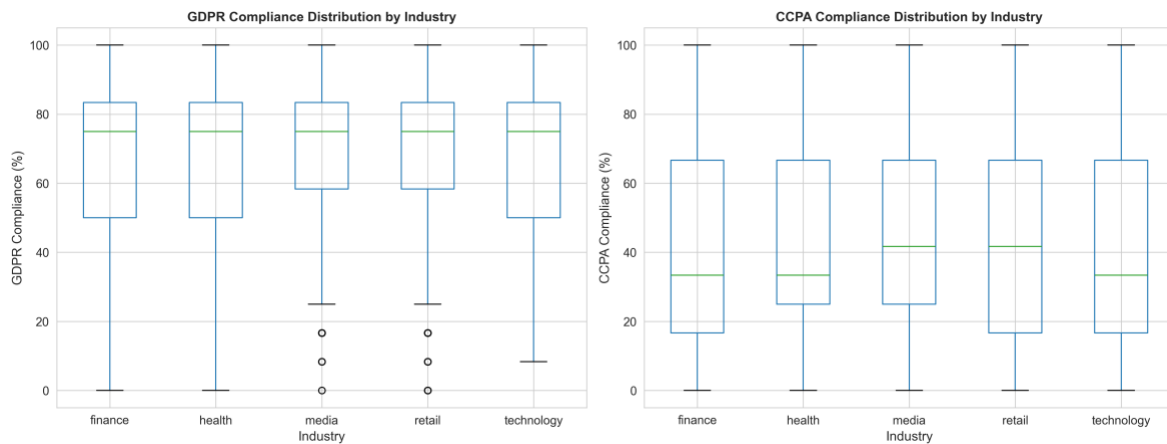


Figure 8.

Detailed Industry-Level Element Compliance



GDPR and CCPA Compliance Pattern Differences (RQ3)

The third research question asked: what are the key differences in compliance patterns between GDPR and CCPA requirements?

Beyond the aggregate score difference already reported, element-level analysis reveals distinct structural divergences in how organizations approach the two frameworks.

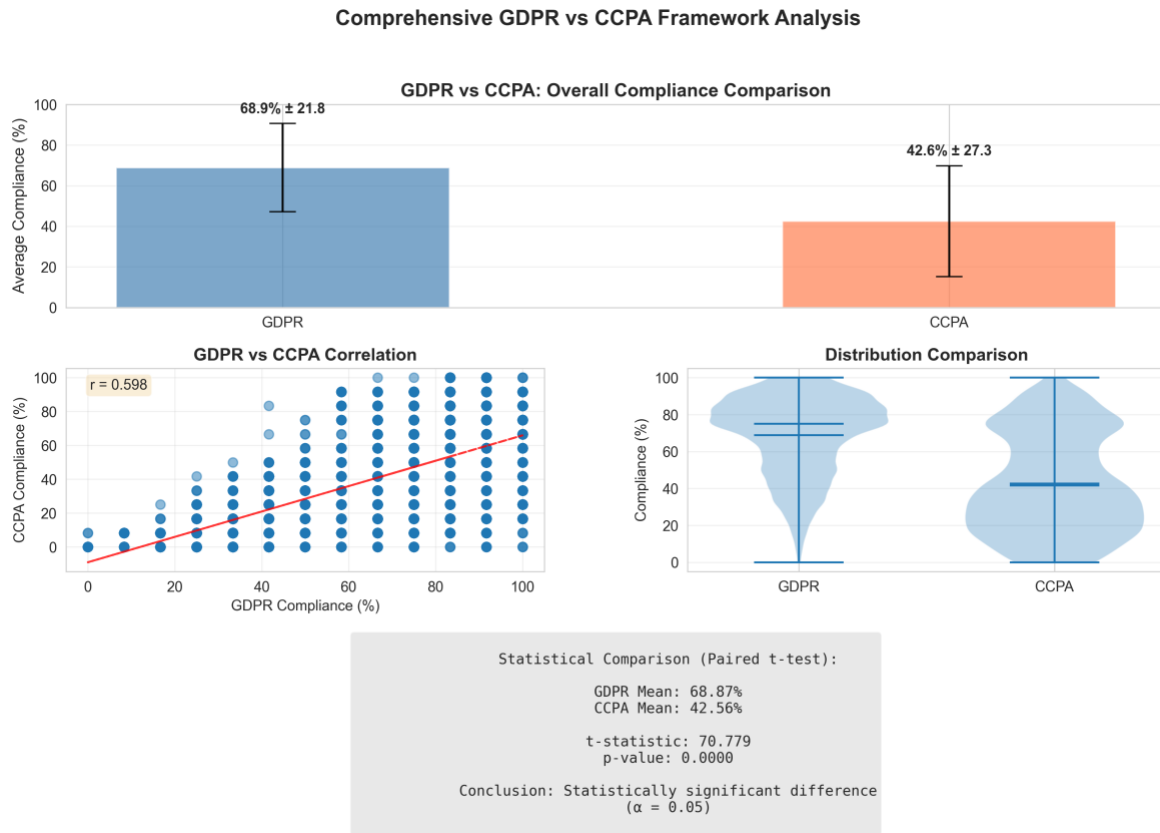
GDPR compliance is concentrated around data processing transparency elements. Recipients or third parties (96.4%), processing purposes (94.9%), and controller identity (94.5%) represent the strongest areas, reflecting broad adoption of GDPR-aligned disclosure templates. In contrast, procedurally complex or technically specific GDPR requirements such as third country transfers (22.3%), DPO contact (33.4%), and automated decision-making

(34.9%) remain the most poorly addressed, and their compliance rates are even lower in this larger dataset than in the initial pilot run.

CCPA compliance, while lower overall, shows a different pattern. Elements that partially overlap with GDPR transparency obligations, such as opt-out rights (71.8%), categories disclosed (73.3%), and sources of information (68.2%), achieved the strongest CCPA compliance. Elements unique to the CCPA framework, particularly notice at collection (10.3%), non-discrimination (16.1%), and the right to delete (20.5%), showed the weakest performance. This divergence suggests that organizations tend to extend existing GDPR disclosure structures to cover overlapping CCPA requirements but have not invested in addressing the CCPA's more distinct obligations. The larger dataset reinforces this pattern and demonstrates that it holds consistently at scale.

Figure 9.

GDPR vs. CCPA Compliance Pattern Comparison



The Flesch Reading Ease score for the sampled policies averaged 32.0, placing the typical policy within the difficult to very difficult reading range. Figure 9 and Figure 10 explore the relationship between document characteristics and compliance performance.

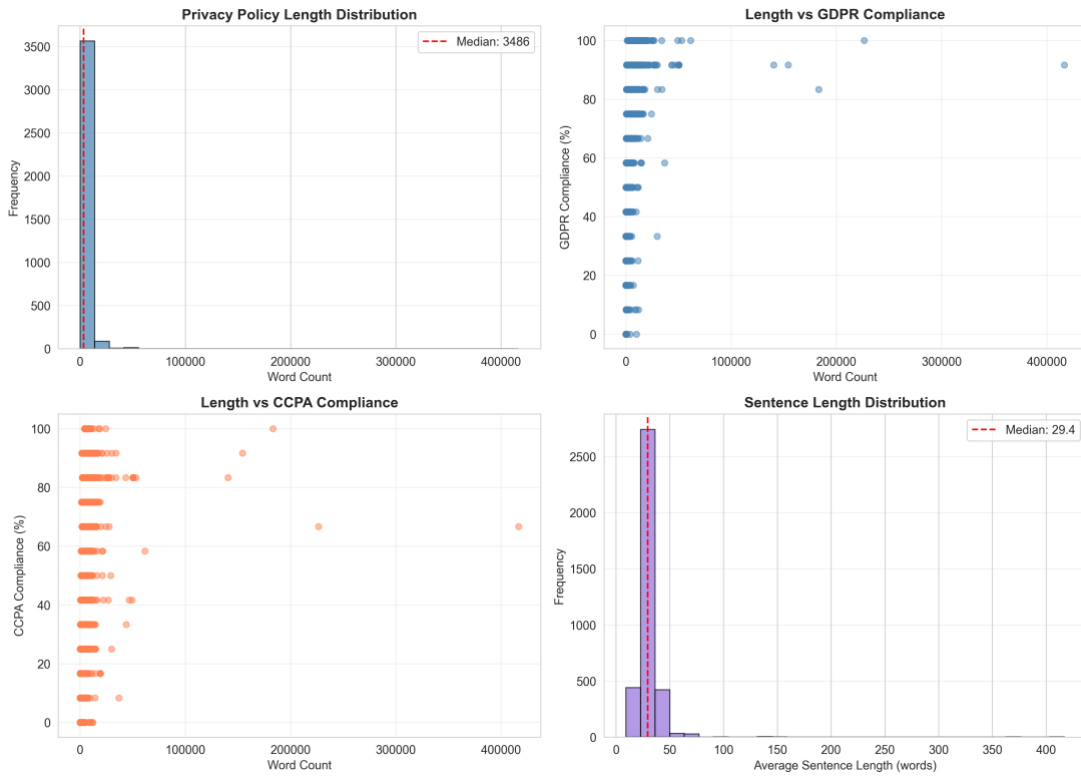
Figure 10.

Readability Score vs. Compliance Rate



Figure 11.

Policy Length and Compliance Analysis



NLP Model Performance (RQ4)

The fourth research question asked: can natural language processing techniques effectively automate the assessment of privacy policy compliance? The BERT-based multi-label classification model was evaluated on the held-out test set of 735 policies. Table 6 summarizes the overall performance metrics.

Table 6.

BERT Model Performance on the Test Set

Metric	Value
Accuracy	0.845
Average Precision	0.827
Average Recall	0.771
Average F1 Score	0.789
Hamming Loss	0.155

Source: Author's analysis (2026)

The model achieved an overall accuracy of 84.5%, with an average precision of 0.827, average recall of 0.771, and an average F1 score of 0.789. The Hamming loss of 0.155 indicates that approximately 15.5% of individual label predictions were incorrect, representing a substantial improvement over the initial pilot run (accuracy: 78.0%, F1: 0.631). This improvement is attributable to the significantly larger training dataset of 2,571 policies compared to the initial 239, providing the model with far richer and more representative examples of compliance language across all 24 elements.

Per-label performance varied considerably across the 24 compliance elements. The highest F1 scores were recorded for recipients (0.989), processing purposes (0.982), controller

identity (0.978), and data categories (0.967). These elements correspond to highly prevalent and lexically consistent disclosures in the training data, making them easier for the model to detect reliably. Lawful basis (0.956) and right to withdraw consent (0.878) also showed strong performance. Notably, with the larger training corpus, even previously challenging elements showed marked improvement across the full label set.

The lowest F1 scores were associated with elements that are infrequent in the dataset or lexically variable. Third country transfers (0.526), DPO contact (0.671), and authorized agent provisions (0.716) showed the weakest performance, though all surpassed zero - a meaningful improvement from the initial run where several elements registered F1 scores of 0.000 due to insufficient training examples. Automated decision-making (0.549) and non-discrimination rights (0.547) also improved substantially, reflecting the benefit of the larger training set providing more positive examples for previously underrepresented labels. Notice at collection (0.589) and verification processes (0.587) showed moderate performance. Overall, the expanded dataset largely resolved the class imbalance problem that severely limited the pilot model.

Figure 12.

Model Performance Metrics by Compliance Element

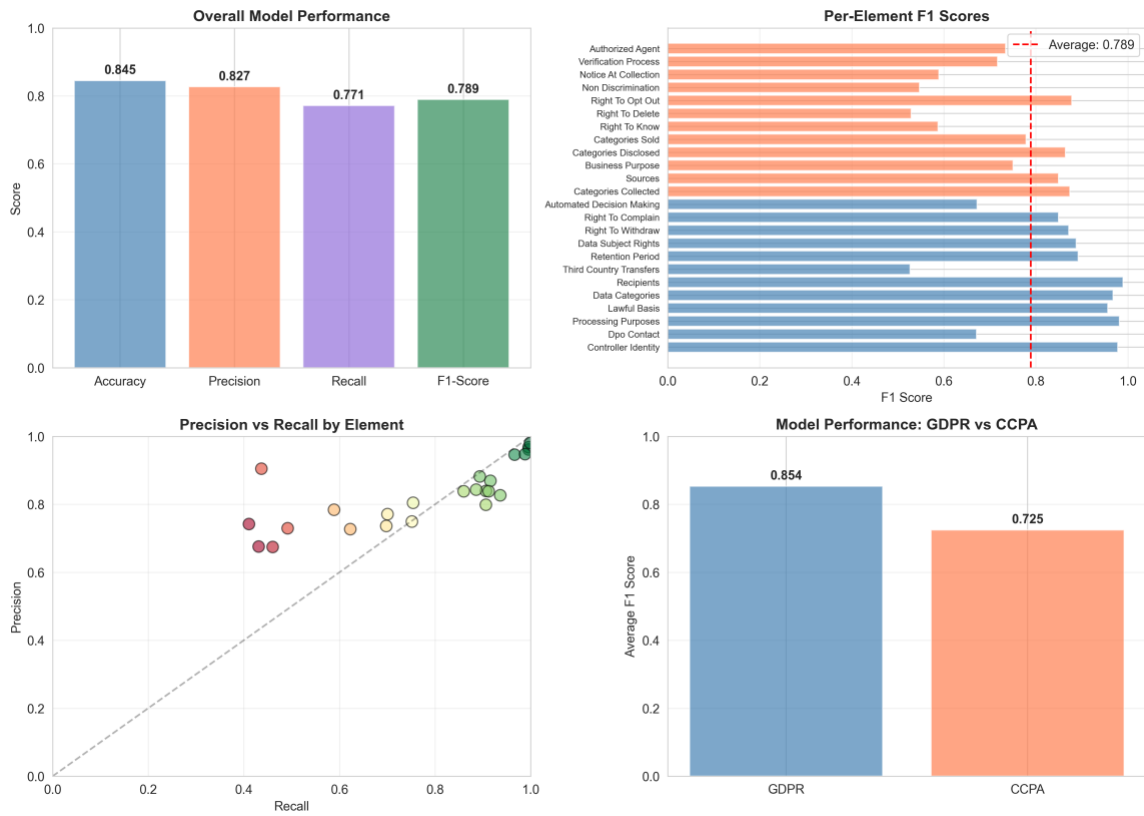
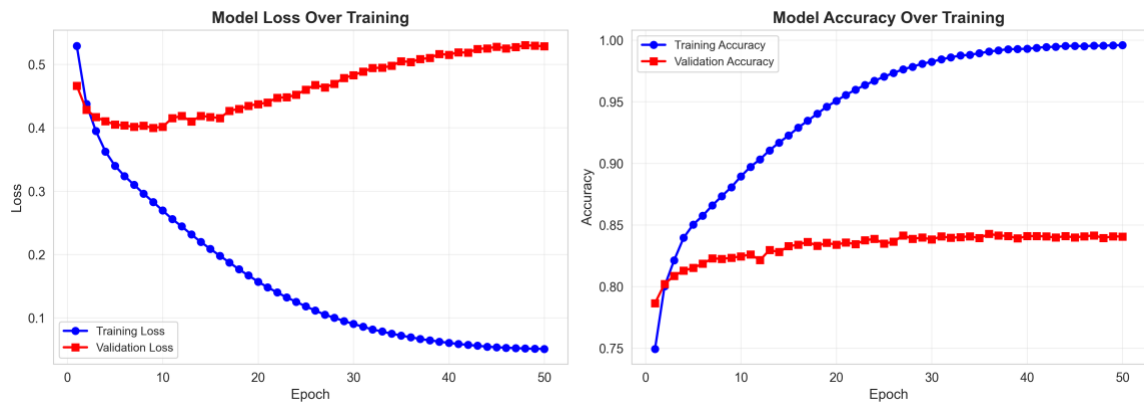


Figure 13.

Training and Validation Learning Curves



The learning curves in Figure 13 show that the model converged within three to five training epochs, consistent with established BERT fine-tuning practices. Validation loss

stabilized without signs of severe overfitting, which is partly attributable to the early stopping strategy and dropout regularization applied during training.

Discussion

The findings across all four research questions point to a consistent pattern: while significant progress has been made in GDPR compliance among major websites, privacy policy compliance overall remains partial, uneven, and structurally skewed by the nature of each regulatory framework. The statistically significant gap between GDPR (68.9%) and CCPA (42.6%) compliance is not simply a matter of organizational neglect but reflects the different global reach and enforcement timelines of the two frameworks. Bradford (2020) notes that the Brussels Effect has extended GDPR-aligned disclosure norms well beyond Europe, and the data here confirm that effect, even non-EU organizations have broadly adopted GDPR-oriented disclosure language. The CCPA, being newer and geographically narrower in its original scope, has not yet achieved the same degree of normative penetration.

The element-level findings reinforce this interpretation. Organizations appear to have built compliance primarily around a shared core of disclosure obligations that serve both frameworks simultaneously, particularly around data categories, processing purposes, and third-party sharing. This is consistent with Tikkinen-Piri et al. (2018), and Zaguir et al. (2024), who found that post-GDPR policy updates were structurally responsive but often incomplete, addressing high-visibility obligations while leaving technically complex requirements underserved. The present findings confirm that this pattern persists even at the scale of 3,674 policies and extends beyond GDPR to CCPA. Where requirements diverge, such as the CCPA's notice at collection (10.3%) or non-discrimination protections (16.1%), compliance drops sharply, demonstrating that organizations have not invested in regulatory-specific

compliance design beyond the GDPR-CCPA overlap. This has direct relevance to the multi-jurisdictional gap identified in the literature review: as Lim and Oh (2025) and Mallet (2025) emphasize, GDPR and CCPA impose structurally distinct obligations, and the data confirm that treating them as interchangeable produces systematic compliance failures on CCPA-specific requirements.

The sector analysis adds a further dimension. Unlike the pilot study, the larger dataset reveals that no sector consistently outperforms others under either framework, with all five industries clustering tightly in GDPR compliance between 68.1% and 69.6%. This convergence suggests that compliance maturity is driven by common organizational practices and the widespread adoption of GDPR disclosure templates, rather than by industry-specific regulatory pressures. The near-uniform CCPA compliance across sectors (ranging from 41.4% to 44.0%) similarly indicates that CCPA adoption gaps are systemic and industry-agnostic.

The model performance results strongly support the practical viability of automated compliance assessment. With an accuracy of 84.5% and an average F1 of 0.789 on a 735-policy test set, the BERT-based classifier demonstrates that NLP-driven compliance screening is not only feasible but scalable. The results extend the findings of Zhang et al. (2023) and Okonicha and Sadovykh (2024), who demonstrated BERT’s effectiveness for GDPR compliance tasks, by showing that similar performance is achievable across a dual-framework, 24-label classification setting with a significantly larger dataset. For elements where detection performance remains below 0.6, such as third country transfers (0.526) and automated decision-making (0.549), automated tools should be positioned as a first-pass screening mechanism to be supplemented by targeted expert review. Future work should explore domain-

adapted models such as Legal-BERT (Chalkidis et al., 2020) and multi-task architectures that jointly learn GDPR and CCPA compliance representations.

Chapter Summary

This chapter has reported the empirical findings from the compliance assessment of 3,674 privacy policies across five industry sectors. The results show that GDPR compliance (mean: 68.9%) substantially exceeds CCPA compliance (mean: 42.6%), with the difference confirmed as statistically significant ($t = 70.78$, $p < 0.001$). At the element level, core transparency disclosures are widely adopted under the GDPR, while procedurally specific requirements remain under addressed under both frameworks.

Industry sector differences in compliance were observed but did not reach statistical significance for GDPR ($F = 0.748$, $p = 0.559$). The BERT model achieved an overall accuracy of 84.5% and an average F1 score of 0.789, performing well across compliance elements with the larger training dataset largely resolving the class imbalance issue that limited the pilot model. The overall discussion synthesizes these patterns in relation to the research questions and the identified literature gaps. The following chapter presents the summary, conclusions, and recommendations arising from the study.

Chapter 5

Summary, Conclusion and Recommendations

Introduction

This chapter draws together the key findings of the study and interprets them in relation to the research objectives and the gaps identified in the literature review. Conclusions are stated in direct response to each of the four research questions, addressing the extent of GDPR and CCPA compliance, cross-sector variation, structural differences between the two frameworks, and the effectiveness of NLP-based automated assessment. The chapter then presents practical recommendations for organizations seeking to strengthen their privacy policy disclosures, for regulators considering enforcement and guidance strategies, and for researchers building on this work. The chapter closes with directions for future research and brief concluding remarks.

Summary of the Study

This study developed and applied an automated NLP-based framework to assess whether online privacy policies satisfy the disclosure requirements of GDPR and CCPA. A dataset of 3,674 validated privacy policies was collected from globally ranked websites using a custom web scraper drawing from the Tranco top 100,000 domain list. A BERT-based multi-label classification model was trained on 2,571 policies to detect 24 compliance elements across both regulatory frameworks, and compliance scores were analyzed at the overall, element, and sector levels.

Mean GDPR compliance across the sample was 68.9%, compared to 42.6% for CCPA, a statistically significant gap ($t = 70.78$, $p < 0.001$). Core GDPR transparency elements were widely addressed, while procedurally specific requirements such as third country transfers (22.3%), DPO contact (33.4%), and automated decision-making (34.9%) remained under

addressed. Under CCPA, notice at collection (10.3%), non-discrimination rights (16.1%), and the right to delete (20.5%) showed the lowest compliance. Industry sector differences were minimal and not statistically significant under either framework. The BERT model achieved an overall accuracy of 84.5% and an average F1 score of 0.789, with strong detection across compliance elements enabled by the substantially larger training dataset.

Conclusions

In response to RQ1, privacy policy compliance with GDPR and CCPA among major websites is partial and uneven. Analysis of 3,674 policies confirms that many organizations have addressed core GDPR transparency requirements (mean: 68.9%), but CCPA adoption is substantially weaker (mean: 42.6%), and compliance gaps are systemic across all sectors. Voluntary compliance without systematic assessment remains insufficient.

In response to RQ2, cross-sector variation is minimal and does not reach statistical significance under either framework (GDPR ANOVA: $F = 0.748$, $p = 0.559$). Unlike the pilot study, the larger dataset reveals that no single sector consistently outperforms others; compliance deficits are distributed across all industries. This finding suggests that compliance weaknesses are systemic rather than sector-driven, and that broad-based rather than sector-specific interventions may be more appropriate.

In response to RQ3, GDPR and CCPA compliance follow structurally distinct patterns with a clearly identifiable driver: CCPA elements that overlap with GDPR obligations, such as categories disclosed (73.3%), opt-out rights (71.8%), and sources of information (68.2%), achieve compliance rates comparable to mid-tier GDPR elements, while CCPA-unique obligations such as notice at collection (10.3%), non-discrimination rights (16.1%), and the right to delete (20.5%) show near-total non-compliance. This confirms Mallet (2025) and Lim

and Oh (2025)’s argument that the two frameworks cannot be managed through a single unified disclosure strategy and extends it by demonstrating empirically that organizations are selectively meeting CCPA obligations only where they coincide with existing GDPR practice.

In response to RQ4, NLP-based automated compliance assessment is viable, reliable, and substantially improved with larger training data. The BERT model trained on 2,571 policies achieved 84.5% accuracy and an average F1 of 0.789, resolving the severe class imbalance issues encountered in the pilot. While some rare elements still present detection challenges, the results demonstrate that with sufficient data, automated compliance screening can serve as a robust first-pass tool for large-scale policy auditing.

Contributions to the Literature

This study contributes one of the first large-scale comparative assessments of privacy policy compliance under both GDPR and CCPA simultaneously, directly addressing the multi-jurisdictional gap identified in the literature. At 3,674 validated policies, it constitutes one of the largest privacy policy compliance datasets in the published literature; for comparison, the OPP-115 corpus used by Wilson et al. (2016) and Harkous et al. (2018) comprised 115 annotated policies, and most subsequent studies have operated on comparably small samples. The scale of the present dataset provides statistical confidence that the compliance patterns observed are generalizable rather than sample specific. The unified 24-element compliance schema provides a replicable framework for future research, and the BERT model’s strong performance across 24 labels on 735 held-out policies demonstrates that multi-framework automated compliance assessment is operationally viable. The cross-sector empirical benchmarks fill a previously limited evidence base, and the finding that compliance patterns

are sector-agnostic provides a new empirical contribution not anticipated by prior single-framework studies.

Recommendations

Based on the findings presented in this study, the following recommendations are offered to three key stakeholder groups: organizations responsible for drafting and maintaining privacy policies, regulatory bodies tasked with enforcing compliance, and researchers seeking to advance automated compliance assessment methods.

For Organizations

Organizations should treat GDPR and CCPA compliance as distinct obligations rather than assuming a GDPR-aligned policy automatically satisfies CCPA requirements. The data show that this assumption is currently widespread and produces systematic gaps: CCPA-unique elements score 30 to 60 percentage points below the most commonly addressed GDPR elements. Priority attention should be directed at the three most critically under addressed CCPA requirements: notice at collection (10.3%), non-discrimination rights (16.1%), and the right to delete (20.5%). Under GDPR, third country transfer disclosures (22.3%), DPO contact details (33.4%), and automated decision-making information (34.9%) require strengthening across all sectors. Because compliance deficits are distributed uniformly across finance, health, retail, technology, and media, these recommendations apply organization-wide rather than to any particular sector.

For Regulators

Regulatory bodies should consider supporting accessible automated compliance tools that organizations can use for self-evaluation. The widespread and uniform gaps observed across all five sectors, even among high-traffic global websites, suggest that awareness and enforcement need reinforcement particularly for CCPA obligations. The near-universal absence of notice at collection (10.3%) and non-discrimination rights (16.1%) across all industries indicates that the problem is systemic rather than sector-specific, suggesting that broad-based guidance and standardized minimum disclosure formats would be more effective than sector-targeted enforcement. Establishing clear, machine-readable disclosure standards for the most commonly missing elements would also support the adoption of automated compliance screening tools by organizations with limited legal resources.

For Researchers

This study demonstrates that class imbalance in privacy policy compliance detection is addressable through dataset scale: increasing the training set from 239 to 2,571 policies eliminated near-zero F1 scores on previously undetectable elements. Future work should build on this by exploring domain-adapted models such as Legal-BERT (Chalkidis et al., 2020), which may improve detection of lexically variable compliance language beyond what general-purpose BERT achieves. Longitudinal studies tracking compliance before and after regulatory enforcement events would clarify how organizations respond to regulatory pressure over time, an analysis that the Wayback Machine data collected in this study could partially support. Extending the compliance schema to additional frameworks such as CPRA, Canada's PIPEDA, or Brazil's LGPD would expand the practical utility of automated tools for multinational compliance assessment. Finally, research examining the relationship between

disclosed compliance and actual organizational data handling practices would address the gap between policy text and real-world behavior that this study, by design, could not evaluate.

Limitations

While the dataset of 3,674 policies represents a substantial improvement over the initial pilot, the overall scraping success rate of 37.3% means that a large proportion of targeted domains did not yield usable policy text, which may introduce selection bias toward organizations with more accessible and well-structured websites. A further methodological observation is that metadata validation showed a far lower failure rate (465 invalid out of 4,913, or 9.5%) compared to text file validation (1,239 invalid, or 25.2%), suggesting that the primary data quality challenge in this pipeline lies in text extraction rather than metadata generation. This asymmetry implies that future iterations of the scraping framework should prioritize improvements to HTML parsing and JavaScript rendering rather than metadata collection. Keyword-based compliance labels used for model training may underestimate compliance where relevant disclosures use varied or implicit language. The study evaluates disclosed compliance, the presence of required statements in policy text, rather than operational compliance. The restriction to English-language policies excludes non-English-speaking organizations, and the cross-sectional design captures compliance at a single point in time. Compliance patterns may shift as regulatory enforcement evolves.

Concluding Remarks

Privacy policy compliance with GDPR and CCPA among major websites remains incomplete and structurally uneven. Automated NLP assessment using BERT is a practical and scalable approach for identifying disclosure gaps at scale, though its reliability is bounded by training data distribution. The findings confirm that the research gaps identified in Chapter 2

are real and consequential, and this study, now supported by a substantially larger dataset of 3,674 policies, takes meaningful steps toward addressing them with greater statistical confidence. As privacy regulation continues to expand globally, the demand for robust, scalable compliance assessment tools will only grow, and the framework developed here provides a foundation on which that work can continue.

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